

HARSH VARDHAN MISHRA

[GitHub](#) | [LinkedIn](#) | harshvmishra123@gmail.com | +91 6387557185

WORK EXPERIENCE

Associate Data Scientist - Partex.AI

June 2025 - Present

- Built an end-to-end **Bank Reconciliation System (BRS)** using a multi-pass matching pipeline (amount/date, UTR, cheque, and fuzzy narrative matching), achieving **81% accuracy**.
- Developed a **LightGBM-based** multi-class desk allocation model on **850K** financial transactions, achieving **87.8% overall accuracy** and **96.9%** on high-confidence auto-routed predictions; deployed via **FastAPI**.
- Built a **CLIP-style** multimodal model using **GIN** and **ESM-2**, trained on **2.3M** drug-protein pairs, achieving **85% Hit@10** under scaffold- and homology-based **OOD evaluation**.
- Optimized proteome-wide inference with **FAISS** and cached **embeddings** across **3,100+** proteins (**0.3s** latency/drug), and developed a deterministic pipeline retrieving evidence from external **APIs**.
- Architected a **Graph-Based AI** system for **drug discovery**, combining the capability of biomedical **knowledge graphs** to model drug-disease relationships at scale (**20M+ connections**).
- Handled real-world data complexity by designing a reliable data integration pipeline and stabilizing large-scale **GNN training** on high-dimensional biomedical data.
- Built **DGSSynADR**, a deep learning system that predicts cancer drug combinations across **80+ cell lines**, reducing millions of candidates to a shortlist for experimental validation.
- Improved performance and scalability (**0.58 PCC**, **85% accuracy**) through **ensemble modeling** and efficient training, while enabling predictions for new drugs from chemical structure.

Data Science Consultant - Partex.AI

July 2024 - June 2025

- Developed an advanced **Q&A ChatBot** with contextual history and memory management, enhancing knowledge retrieval **efficiency by 80%** for R&D teams.
- Implemented **Retrieval Augmented Generation (RAG)**, cutting response time by **50%** and **boosting data retrieval accuracy by 75%**.
- Optimized **scalable data access** using **DaaS** and **FAISS**, achieving **30%** faster query processing through efficient **vector embeddings**.

TECHNICAL SKILLS

Languages: Python

Libraries: NumPy, Pandas, Scikit-learn, Tensorflow, PyTorch, DGL (Deep Graph Library), FastAPI .

Technologies/Frameworks: Machine learning, Deep Learning, LSTM, Transformers, Knowledge-Graph, Generative AI, Prompt Engineering, Langchain, LangGraph, RAG, MCP, Hybrid Search.

Developer/AI Tools: Jupyter, VSCode, Cursor AI, Git.

EDUCATION

BCA Data Science at **SRM University, Kattankulathur**

2021 - 2024